

In the middle of the bridge

W24 (2024) — English translation

In this problem, we study electrical circuits containing a nonlinear element, the current-voltage characteristic of which is shown in Fig. 1. The current-voltage characteristic of a nonlinear element is described by two sections: $I = I_0 \cdot (U/U_0)$ at $U \leq U_0$, where $I_0 = 10$ mA and $U_0 = 5$ B. $I = I_0 = \text{const}$ at $U \geq U_0$.

In this problem, we study electrical circuits containing a nonlinear element, the current-voltage characteristic of which is shown in Fig. 1.

The current-voltage characteristic of a nonlinear element is described by two sections:

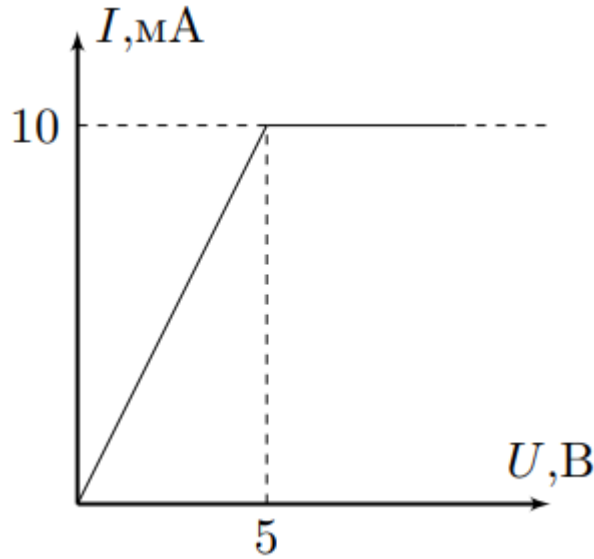


Fig. 1.

Part A. DC Circuit (3.0 points)

Figure 2 shows a diagram of an electrical circuit consisting of resistors with resistances R , $2R$, $4R$, $6R$ and $8R$, an ideal constant voltage source $\mathcal{E}$ and the described nonlinear element. This part of the problem is devoted to studying the dependence of the current through a nonlinear element depending on the source voltage.

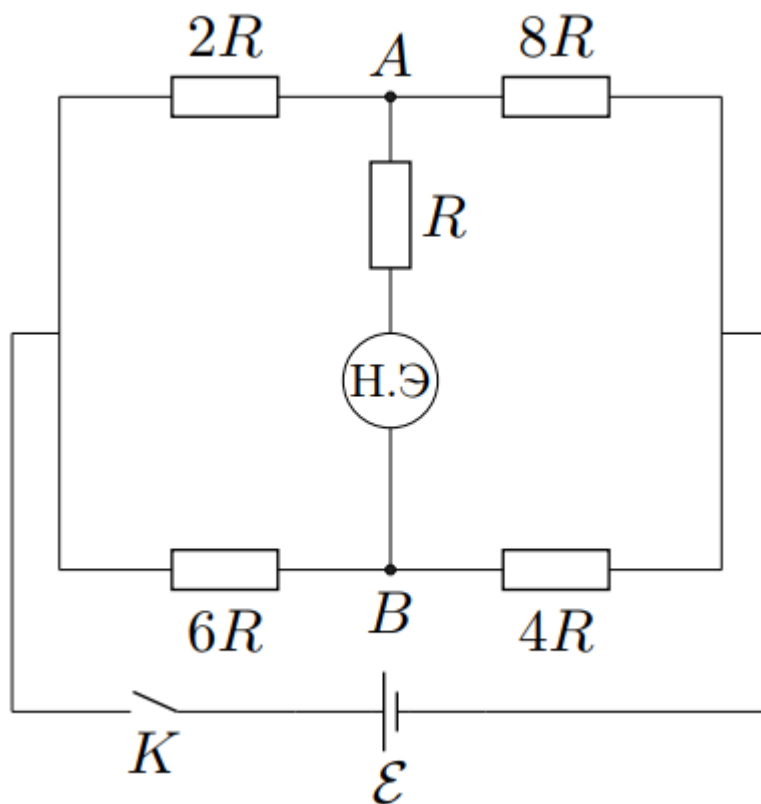


Fig. 2.

- A1** Since without a section of the circuit between nodes A and B the electrical circuit consists entirely of constant voltage sources and resistors, the dependence of the current I_{AB} flowing from node A to node B on the voltage $U_{AB} = \varphi_A - \varphi_B$ in this section of the circuit can be represented in a form that does not depend on the element connected between nodes A and B : **1.50**

$$I_{AB} = \frac{\mathcal{E}_0 - U_{AB}}{r_0}.$$

Определите $\mathcal{E}_0$ и r_0 . Ответы выразите через $\mathcal{E}$ и R .

- A2** Obtain the dependence of the current strength I_{AB} on the value $\mathcal{E}$. Express your answer using $\mathcal{E}$, R , U_0 and I_0 . **0.50**

- A3** Obtain the dependence of the power $P_{\text{H.Э}}$ released on the nonlinear element on the value $\mathcal{E}$. Express your answer using $\mathcal{E}$, R , U_0 and I_0 . **0.50**

- A4** Construct a high-quality graph of the dependence $P_{\text{H.Э}}(\mathcal{E})$. Indicate all the characteristic points on the graph and indicate their corresponding values. **0.50**

Part B. Electric Circuit Containing a Capacitor (5.0 points)

The figure shows an electrical circuit consisting of an ideal voltage source $\mathcal{E}$, a resistor $R = 500 \text{ Ом}$, a capacitor with a capacitance $C = 100 \text{ мкФ}$ and a nonlinear element connected in series, as well as a switch K . Initially, the key is open and the capacitor is not charged. At the initial time $t = 0$ the key is closed.

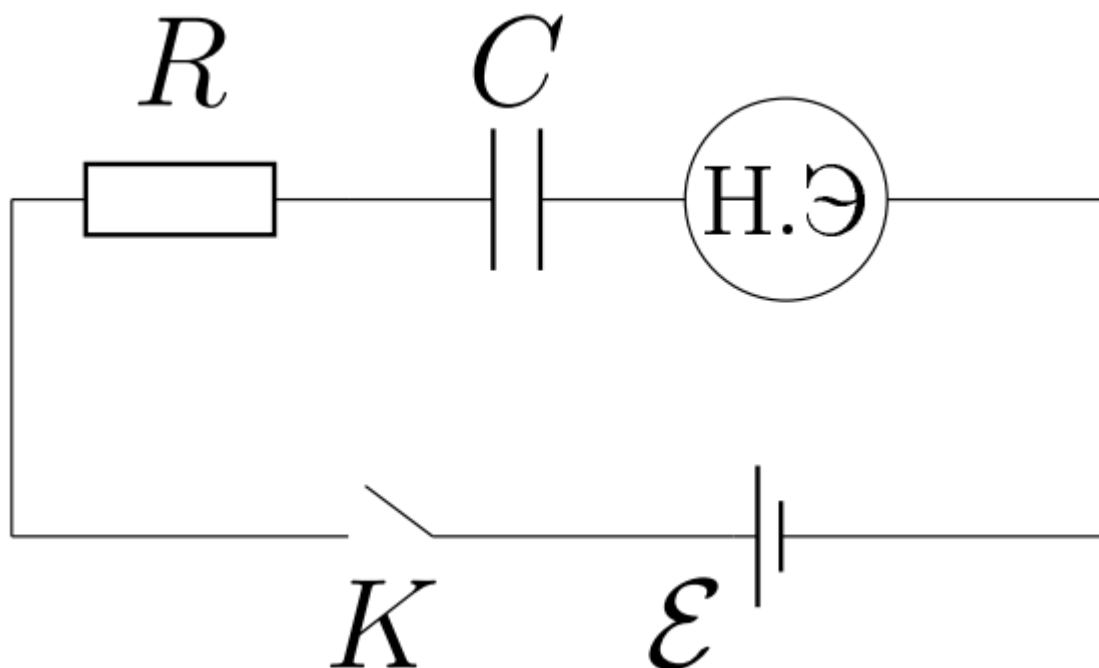


Fig. 3.

Let the emf of the source be equal to $\mathcal{E} = 6$ B.

B1 Obtain the dependence of the current $I_C(t)$ through the capacitor on time t . Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 , C and t . **0.80**

B2 What amount of heat $Q_{H.\mathcal{E}}$ will be released on the nonlinear element during the establishment of constant current? Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 and C . Find the numerical value. **0.60**

B3 After what time τ the amount of heat released on the nonlinear element will reach 80% from the value $Q_{H.\mathcal{E}}$? Express your answer in terms of R , U_0 , I_0 , and C . Find the numerical value. **0.60**

Now the emf of the source is $\mathcal{E} = 12$ B.

B4 Obtain the dependence of the current $I_C(t)$ through the capacitor on time t . Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 , C and t . **1.50**

B5 What amount of heat $Q_{H.\mathcal{E}}$ will be released on the nonlinear element during the establishment of constant current? Find the numerical value with an accuracy of at least 1%. **1.50**

Source: <https://pho.rs/p/3553>